Marks - 5

Class Assignment - 1

Q1. Determine whether the following is tautology

(a) $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$

(b) $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$

Q2. Show that $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$ are logically equivalent

Q3. The dual of a compound proposition that contains only the logical operators $\vee$, $\wedge$ and $\neg$ is the compound proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each T by F and each F by T.

(a) Find the dual of $(p \wedge q \wedge r) \vee s$

(b) When does Dual of [~~Cou~~ Compound proposition] = Compound proposition.

Q4. How many different truth tables of compound propositions are there that involve the propositional variables p and q. ?

Q5. Translate these statements into English where $R(x)$ is "x is a rabbit" and $H(x)$ is "x hops". Domain is all animals.

(a) $\forall x (R(x) \rightarrow H(x))$ (b) $\forall x (R(x) \wedge H(x))$

(c) $\exists x (R(x) \rightarrow H(x))$ (d) $\exists x (R(x) \wedge H(x))$

Q6. Translate each of these statements into logical expressions using predicates, quantifiers and logical connectives.

(a) No one is perfect
(b) Not everyone is perfect
(c) All your friends are perfect
(d) Atleast one of your friends is perfect
(e) Everyone is your friend and is perfect
(f) Not everybody is your friend or someone is Not perfect

Q7: Express the Negation of these propositions using quantifiers and then Express the Negation in English.

(a) Some drivers do not obey the speed limit
(b) All Swedish movies are serious.
(c) No one can keep a secret
(d) There is someone in this class who does not have a good attitude.

Q8: True / false:

$$\forall x\, P(x) \vee \forall x\, Q(x) \equiv \forall x\,(P(x) \vee Q(x))$$

Q9. As mentioned in the class, the Notation $\exists!\, x\, P(x)$ denotes: There exists a unique X such that $P(x)$ is true.

If the Domain is all integers, what are Truth Value of

(a) $\exists!\, x\,(x > 1)$ (b) $\exists!\, x\,(x^2 = 1)$
(c) $\exists!\, x\,(x+3 = 2x)$ (d) $\exists!\, x\,(x = x+1)$

10. Let $T(x,y)$ mean that student x likes cuisine y, where
Domain (x): all students at your school.
Domain (y): All cuisines.

Express each of these statements by simple English

(a) $\neg T(\text{Ram}, \text{Japanese})$

(b) $\exists x\, T(x, \text{Korean}) \wedge \forall x\, T(x, \text{Mexican})$

(c) $\exists y\, (T(\text{Ram}, y) \vee T(\text{Rajeev}, y))$

(d) $\forall x \forall z \exists y\, ((x \neq z) \rightarrow \neg(T(x,y) \wedge T(z,y)))$

(e) $\exists x \exists z \forall y\, (T(x,y) \leftrightarrow T(z,y))$

(f) $\forall x \forall z \exists y\, (T(x,y) \leftrightarrow T(z,y))$

11. Rewrite each of these statements so that negation appear only within predicates.

(a) $\neg \exists y \exists x\, P(x,y)$ (b) $\neg \forall x \exists y\, P(x,y)$

(c) $\neg \exists y\, (Q(y) \wedge \forall x \neg R(x,y))$

(d) $\neg \exists y\, (\exists x\, R(x,y) \vee \forall x\, S(x,y))$

(e) $\neg \exists y (\forall x \exists z\, T(x,y,z) \vee \exists x \forall z\, U(x,y,z))$

12 $\forall x\, P(x) \vee \forall x\, Q(x) \equiv \forall x \forall y\, (P(x) \vee Q(y))$
where all quantifiers have the same non empty domain are logically equivalent. [True/false]

13. What is the difference between the Quantificati

(a) $\exists x \forall y\, P(x,y)$ and $\forall y \exists x\, P(x,y)$ where $P(x,y)$ is a predicate ?

(b) Give a Counter Example of a predicate $P(x,y)$ such that $\exists x \forall y\, P(x,y)$ and $\forall y \exists x\, P(x,y)$ have different truth values.

Q14: Show that the argument

" If the earth is flat, then you can sail off the edge of the earth, " " you cannot sail off the edge of the earth

therefore the earth is NOT flat is valid argument or NOT.